

PicoCTF 2026 - Ksecrets

KSECRETS

General Skills

Medium

100 pts 1,715 solves

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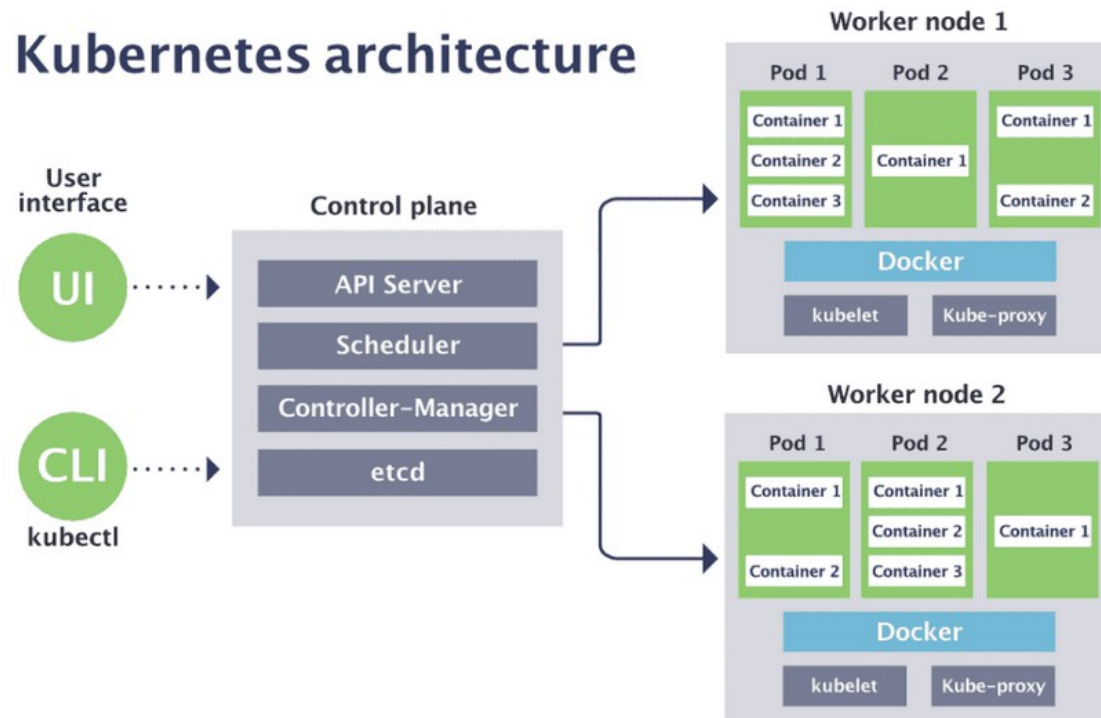
We have a kubernetes cluster setup and flag is in the secrets. You think you can get it?
Please wait for a minute for the application to be configured!
Kubernetes is running at: green-hill.picoctf.net : 60449
You can find your configuration file [here](#).

In this challenge, we're asked to enumerate secrets in a Kubernetes cluster

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Kubernetes Clusters

Kubernetes architecture



Kubernetes clusters are sets of machines (called nodes) that run containerized applications

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Kubernetes Secrets



In a Kubernetes cluster, there are objects called secrets, which store and manage sensitive data, like passwords, SSH keys, and API keys

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Kubernetes Setup

```
theshyhat-academy@webshell:~/ksecrets$ export KUBECONFIG=./kubeconfig  
theshyhat-academy@webshell:~/ksecrets$ sed -i 's|127.0.0.1:6443|green-hill.p  
icoctf.net:52061|g' kubeconfig
```

To interact with the challenge's Kubernetes cluster, we need to setup the environment

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Enumerating Kubernetes Secrets

```
theshyhat-academy@webshell:~/ksecrets$ kubectl get secrets --all-namespaces --insecure-skip-tls-verify > secrets.txt
```

```
theshyhat-academy@webshell:~/ksecrets$ cat secrets.txt
```

	NAMESPACE	NAME	TYPE
DATA	AGE		
	kube-system	chart-values-traefik	helmcharts.helm.cattle.io/values
1	4m46s		
	kube-system	chart-values-traefik-crd	helmcharts.helm.cattle.io/values
0	4m46s		
	kube-system	k3s-serving	kubernetes.io/tls
2	4m48s		
	picoctf	ctf-secret	Opaque

Then we can use the kubectl command to get all of the secrets object locations in the cluster and output them to a file

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Enumerating Kubernetes Secrets

```
theshyhat-academy@webshell:~/ksecrets$ kubectl get secret ctf-secret -n pico  
ctf -o yaml --insecure-skip-tls-verify  
apiVersion: v1  
data:  
  flag: cGljb0NURntrczNjcjM3NV80MW43X3M0ZjNfM2NhZjhjMzF9Cg==  
kind: Secret
```

With the name of the secret and the namespace,
we can access it directly: